Reg. No. | | | | | | | | | | | [18MTAVS202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019
PROBABILITY THEORY AND MODELING AND SIMULATION

Time: 03 Hours **Maximum Marks:100**

Instructions :

i. Answer FIVE full questions. Question 1 and 2 are compulsory.

ii. Statistical tables are to be issued.

iii. Missing data may be chosen.

1. a. Define **10**

i. Dependent and Independent events

ii. Exhaustive Event

iii. Mutually Exclusive Events,

iv. Probability

v. Conditional Probability and explain multiplication rule of Probability.

b. The probability density function $p(x)$ of a variate 'x' is given by the following table. **05**

x	0	1	2	3	4	5	6
$p(x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

For what value of k does this represent a valid probability distribution? Also find the $p(x<4)$, $p(x \geq 5)$ and $p(3<x \leq 6)$.

c. A box contains 2 white and 4 black balls. Another box B contains 5 white and 7 black ball .A ball is transferred from box A to box B. Then a ball is drawn from box B. Find the probability that it is white? **05**

2. a. Write the steps in simulation study. **10**

b. Define model of a system and write the types of the model. **10**

3. a. i. Define Geometric distribution and write its mean and standard deviation. **04**

ii. The probability that a man aged 60 will live up to 70 is 0.65.What is the probability that out of 10 men, now aged 60, at least 7 will live up to 70. **06**

b. In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and standard deviation of the distribution. **10**

OR

4 a. i. Define uniform distribution. Write its mean and standard deviation. **04**

ii. Buses in metropolitan service arrive at a particular station with frequency of 15 minutes starting from 7AM. A person comes to the station every day at a time uniformly between 7 and 7.30AM. What is the probability that on any day **06**

I. He has to wait less than 5 minutes

II. He has to wait for at least 12 minutes.

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b. The service time of machine is an exponential distribution with an average time 2 hours. **10**

i. What is the probability that a randomly selected machine is repaired before 2 hours

ii. What is the probability that the machine repair time exceeds $2\frac{1}{2}$ hours given that it is not repaired in 2 hours.

5. **a.** Generate 15 random numbers by linear congruential method with $x_0 = 6$ and $x_n = [12 + 21x_{n-1}, mod 25]$ **10**

b. If X is a Bernoulli's random variable with probability of success as 0.6 then generate 16 Bernoulli random variables, for the following random numbers 0.78,0.952,0.905,0.82,0.873,0.083,0.918,0.136,0.149,0.433,0.913,0.999,0.077, 0.216,0.972,0.367 **10**

OR

6. **a.** The number of accidental deaths in a city is a Poisson random variable with an average number of deaths 3.7 per day .Model this problem simulating for a random week(Mon-Sat) **10**

b. **i.** Describe the process of generation of discrete random variables. **05**

ii. The statistics of timings taken over a month of arrival of a train at a particular station is as follows **05**

T	9am	9:05am	9:10am	9:15am
P	6days	12days	7day	5days

Simulate this problem for 7 days starting from Monday by taking the random numbers 0.327, 0.543,0.083,0.911,0.964,0.177,0.762.

7. **a.** Test the uniformity at 5% and 10% level of significance using Kolmogorov Smirnov test of the following random numbers 0.94,0.08,0.35,0.24,0.05,0.58,0.57,0.74,0.87,0.13 **10**

b. The inter arrival time X is given as 0, 3, 5,7,16 in minutes. The service time is given as 2, 5 2, 4, 4 in minutes. Draw the graph of the queue system and find the average time spent by any customer in the system. **10**

OR

8. **a.** A box contains 8 balls of which 5 are black and 3 are white. A person picks 3 balls at a time, replaces and repeat the experiment for five times. Simulate the problem for number of black balls in each experiments. Use following random numbers for the simulation 0.455, 0.163,0.763,0.549,0.02. **10**

b. Check the following 20 random numbers for uniformity using Chi Square test by preparing a distribution table consisting 4 classes at 10% LOS 0.73,0.18,0.0.26,0.47,0.74,0.91,0.76,0.93,0.66,0.37,0.42,0.77,0.19,0.58,0.03,0.96,0.15,0.23,0.07,0.43 **10**